

Mohammed El Kassoiri

Junior Data Scientist | Machine Learning & AI Engineer

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PROFESSIONAL SUMMARY

Junior Data Scientist and Machine Learning Engineer with hands-on experience in deep learning, computer vision, geospatial AI, and natural language processing (NLP). Experienced in designing and developing end-to-end AI pipelines, covering the complete machine learning lifecycle from data acquisition, preprocessing, annotation, feature engineering, and transfer learning to model development, evaluation, deployment, and API integration. Strong background in building intelligent AI solutions and retrieval-augmented generation (RAG) systems, with a passion for developing scalable, production-ready applications that address complex scientific and real-world challenges.

EDUCATION

Diplôme d'Ingénieur en Science des Données, Big Data et Intelligence Artificielle (Engineering Degree) 2023 - 2026
École Nationale Supérieure de l'Intelligence Artificielle et des Sciences des Données (ENSIASD), Morocco

CPGE - Physics & Engineering Sciences (PSI) 2021 - 2023
Lycée Ibn Timiya, Marrakech, Morocco

PROFESSIONAL EXPERIENCE

Intern - Automatic Agricultural Plot Delineation (PFE) Feb 2026 - Jun 2026
Pôle Digital de l'Agriculture, de la Forêt et Observatoire de la Sécheresse, Morocco

- Designed an end-to-end satellite data pipeline, including shapefile ingestion, QGIS processing, rasterization, tiling, and RGB/mask dataset generation, reducing manual preprocessing time by approximately 60%.
- Developed an enhanced DeepLabV3+ dual-task segmentation model with auxiliary boundary supervision for accurate agricultural parcel delineation across multi-source satellite imagery.
- Integrated Sentinel-2 imagery via Google Earth Engine (GEE) and vegetation indices (NDVI, NDRE) to improve multi-scale generalization across diverse agricultural regions.
- Evaluated the model using real cadastral data provided by the ANCFCC, achieving an 83.76% geospatial IoU between the predicted parcel boundaries and the official cadastral reference.

Data Science & Machine Learning Intern Aug 2025 - Oct 2025
XAI Morocco

- Contributed to the X-EdTech AI platform, building three production-ready modules: X-Forms (AI form generation), X-Quiz (automated question creation), and X-Table (data visualisation and analysis).

ACADEMIC PROJECTS

Legal AI - MY LAW: Moroccan Legal RAG System 2025
Academic Project - ENSIASD

- Built a multilingual (Arabic/French) retrieval-augmented generation (RAG) system for intelligent question-answering over Moroccan legal texts, deployed via FastAPI and Gradio.
- Implemented optimized semantic search using FAISS vector indexing and SentenceTransformers, achieving sub-second retrieval latency across 10,000+ legal documents and ensured legal accuracy by enforcing strict citation of law articles in every generated response, reducing hallucination risk.
- Technologies: NLP, RAG, FAISS, SentenceTransformers, FastAPI, Gradio, LLMs, Multilingual AI.

Water Stress Mapping - Taroudant Region 2025
Academic Project - ENSIASD

- Designed a semantic segmentation system for agricultural water stress mapping, fusing 16 spectral bands (optical + SAR radar + geospatial indices).
- Implemented an Attention U-Net model achieving 0.92 overall accuracy and 0.50 Dice coefficient on imbalanced satellite imagery, outperforming standard U-Net baselines, using a weak supervision pipeline on Google Earth Engine, eliminating the need for fully annotated ground truth labels.
- Technologies: Remote Sensing, Deep Learning, Attention U-Net, GEE, Weak Supervision, PyTorch.

Robot Arm Object Sorting System 2024
End-of-Year Project - ENSIASD

- Built a real-time object detection and automated colour-based sorting system integrated with a physical robot arm.
- Achieved high-precision colour recognition using OpenCV with <50ms latency for real-time sorting decisions.
- Technologies: Computer Vision, OpenCV, Real-time Systems, Robotics Integration.

TECHNICAL SKILLS

Programming And MLOps	Python, SQL, NumPy, Pandas, SciPy, Matplotlib, Seaborn, FastAPI, Git, Linux, Jupyter, Docker (basic)
ML / Deep Learning	Scikit-learn, PyTorch, TensorFlow, Keras, Hugging Face Transformers, Model Fine-tuning
NLP & LLMs	SentenceTransformers, RAG Systems, FAISS, Information Retrieval, Multilingual NLP
Computer Vision / Geospatial AI	OpenCV, Semantic Segmentation, Object Detection, Remote Sensing, GIS, QGIS, Google Earth Engine, Sentinel-2

LANGUAGES

Arabic Native French Fluent English Fluent